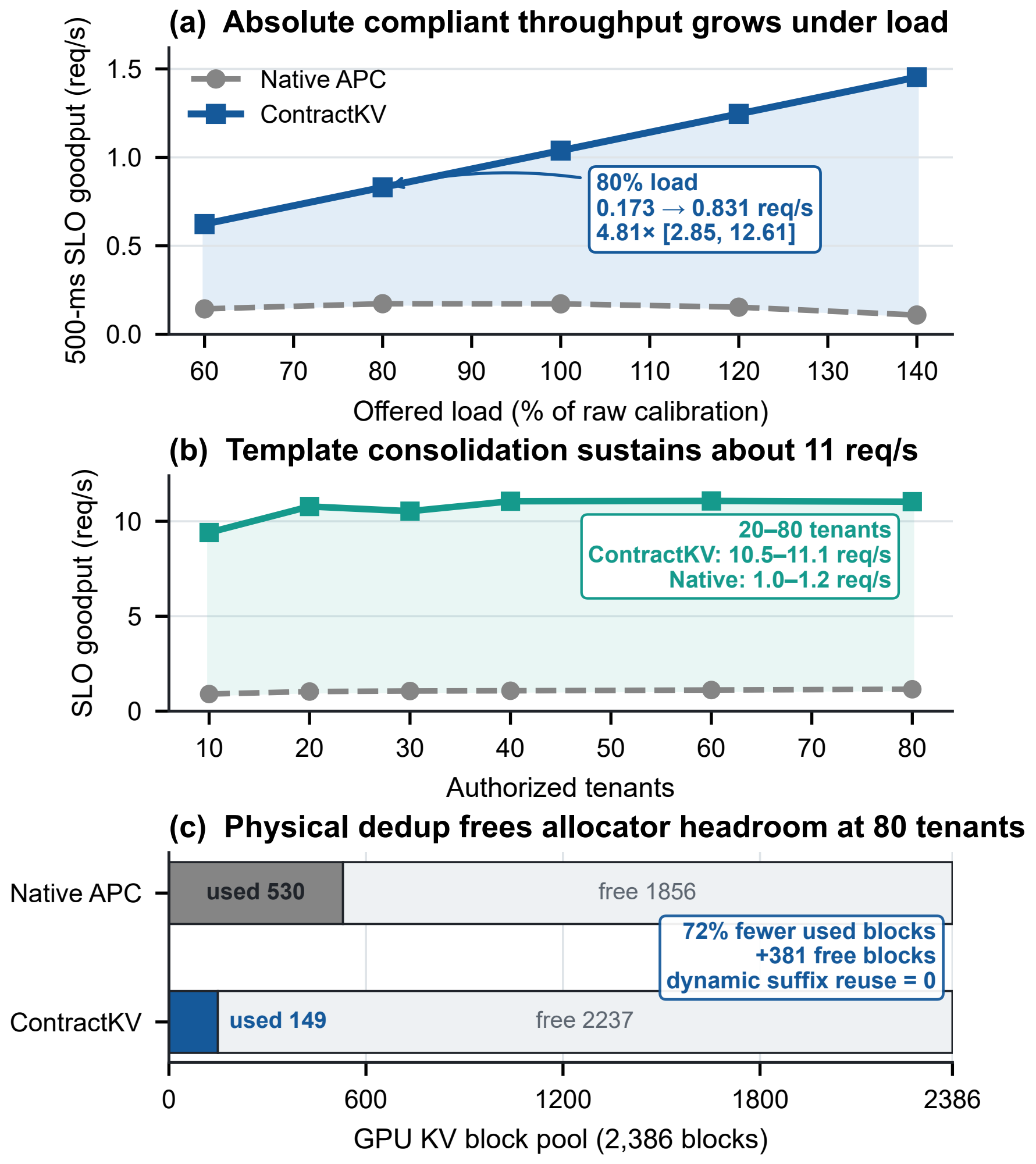

(a) Absolute compliant throughput grows under load
500-ms SLO goodput (req/s)
1.5
1.0
0.5
0.0
Native APC
ContractKV
80% load
0.173 → 0.831 req/s
4.81× [2.85, 12.61]
60
70
80
90
100
110
120
130
140
Offered load (% of raw calibration)
(b) Template consolidation sustains about 11 req/s
SLO goodput (req/s)
10
5
0
20–80 tenants
ContractKV: 10.5–11.1 req/s
Native: 1.0–1.2 req/s
10
20
30
40
50
60
70
80
Authorized tenants
(c) Physical dedup frees allocator headroom at 80 tenants
Native APC
used 530
free 1856
72% fewer used blocks
+381 free blocks
dynamic suffix reuse = 0
ContractKV
used 149
free 2237
0
600
1200
1800
2386
GPU KV block pool (2,386 blocks)